

APPLIED DATABASES PROJECT INNOVATIONS

1. Database Credential Input

Before the application starts, it asks the user to enter the MySQL and Neo4j usernames and passwords with “mysql_user = input()”. Avoiding hardcoded credentials allows the application to run on different machines and VM environments, which may use different database usernames or passwords. It also improves security because database credentials are not stored directly inside the code.

The application also accepts blank usernames or passwords. If a database setup does not require a password, the user can simply press Enter to continue.

2. User Experience Improvement

This is a small usability improvement. The “time.sleep(3)” pauses the application for 3 seconds before returning to the main menu, giving the user enough time to read the results.

Without this pause, the menu would reload immediately, forcing the user to constantly scroll up in the command prompt to view previous results.

3. Top Connected Attendees

This feature displays the 5 attendees with the highest number of connections in the Neo4j database.

The function uses a Neo4j query to: count attendee connections, sort attendees from highest to lowest number of connections and return the top 5 results. It also uses MySQL to retrieve attendee names after the Neo4j query returns attendee IDs.

The final output displays: attendee ID, attendee name and number of connections

4. Suggested Connections

This feature suggests new attendee connections based on mutual connections, similar to a “People You May Know” commonly used in social media platforms.

The query follows a relationship length pattern of 2 connections (second-degree connections): attendee → connection → suggested attendee.

The function: asks the user for an attendee ID, finds attendees directly connected to that attendee, and then finds attendees connected to those attendees. The query excludes: attendees already connected to the current attendee and the own attendee.

Neo4j query returns suggested attendee IDs. Then, using MySQL, it retrieves attendee names for the user.

5. Find Connection Path

This feature finds the shortest connection path between two attendees if it exists.

The function: asks the user for two attendee IDs, validates if both IDs are valid and exist, then searches for the shortest path between them. Using Neo4j’s built-in shortestPath() function where it extracts attendee IDs and displays the connection path to user: 104 -> 120 -> 103.

And finally, using MySQL it retrieves and displays attendee names.